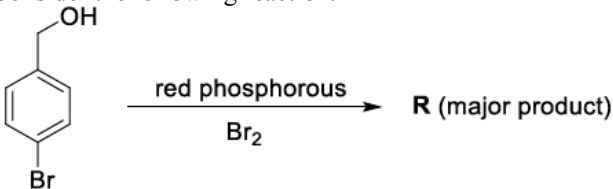


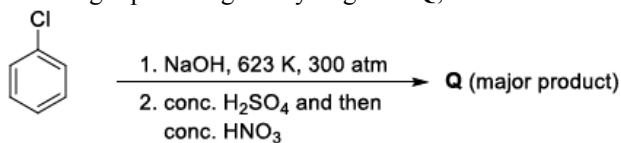
PART III: CHEMISTRY

SECTION 1 (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 **ONLY** if the correct numerical value is entered;
Zero Marks : 0 In all other cases.

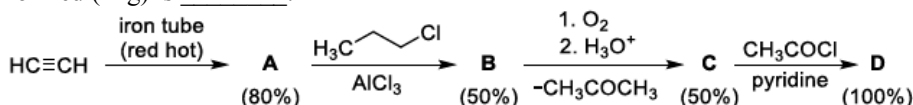
- *Q1. 2 mol of Hg(g) is combusted in a fixed volume bomb calorimeter with excess of O_2 at 298 K and 1 atm into HgO(s) . During the reaction, temperature increases from 298.0 K to 312.8 K. If heat capacity of the bomb calorimeter and enthalpy of formation of Hg(g) are 20.00 kJ K^{-1} and $61.32 \text{ kJ mol}^{-1}$ at 298 K, respectively, the calculated standard molar enthalpy of formation of HgO(s) at 298 K is $X \text{ kJ mol}^{-1}$. The value of $|X|$ is _____.
 [Given: Gas constant $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$]
- Q2. The reduction potential (E^0 , in V) of $\text{MnO}_4^- (\text{aq}) / \text{Mn(s)}$ is _____
 [Given : $E^0_{(\text{MnO}_4^- (\text{aq}) / \text{MnO}_2 (\text{s}))} = 1.68 \text{ V}$; $E^0_{(\text{MnO}_2 (\text{s}) / \text{Mn}^{2+} (\text{aq}))} = 1.21 \text{ V}$; $E^0_{(\text{Mn}^{2+} (\text{aq}) / \text{Mn(s)})} = -1.03 \text{ V}$]
- *Q3. A solution is prepared by mixing 0.01 mol each of H_2CO_3 , NaHCO_3 , Na_2CO_3 , and NaOH in 100 mL of water. $p\text{H}$ of the resulting solution is _____.
 [Given: pK_{a1} and pK_{a2} of H_2CO_3 are 6.37 and 10.32, respectively; $\log 2 = 0.30$]
- *Q4. The treatment of an aqueous solution of 3.74 g of $\text{Cu(NO}_3)_2$ with excess KI results in a brown solution along with the formation of a precipitate. Passing H_2S through this brown solution gives another precipitate **X**. The amount of **X** (in g) is _____.
 [Given: Atomic mass of H = 1, N = 14, O = 16, S = 32, K = 39, Cu = 63, I = 127]
- *Q5. Dissolving 1.24 g of white phosphorous in boiling NaOH solution in an inert atmosphere gives a gas **Q**. The amount of CuSO_4 (in g) required to completely consume the gas **Q** is _____.
 [Given: Atomic mass of H = 1, O = 16, Na = 23, P = 31, S = 32, Cu = 63]
- Q6. Consider the following reaction.
- 
- On estimation of bromine in 1.00 g of **R** using Carius method, the amount of AgBr formed (in g) is _____.
 [Given: Atomic mass of H = 1, C = 12, O = 16, P = 31, Br = 80, Ag = 108].

- Q7. The weight percentage of hydrogen in **Q**, formed in the following reaction sequence, is _____.



[Given: Atomic mass of H = 1, C = 12, N = 14, O = 16, S = 32, Cl = 35]

- Q8. If the reaction sequence given below is carried out with 15 moles of acetylene, the amount of the product **D** formed (in g) is _____.



The yields of **A**, **B**, **C** and **D** are given in parentheses.

[Given: Atomic mass of H = 1, C = 12, O = 16, Cl = 35]

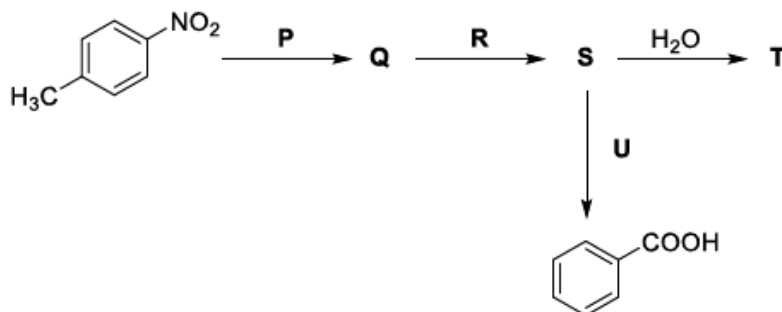
SECTION 2 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
 - Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
 - For each question, choose the option(s) corresponding to (all) the correct answer(s).
 - Answer to each question will be evaluated according to the following marking scheme:
- | | | | |
|----------------|---|----|--|
| Full Mark | : | +4 | If only (all) the correct option(s) is(are) chosen; |
| Partial Marks | : | +3 | If all the four options are correct but ONLY three options are chosen; |
| Partial Marks | : | +2 | If three or more options are correct but ONLY two options are chosen, both of which are correct; |
| Partial Marks | : | +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option; |
| Zero Marks | : | 0 | If unanswered; |
| Negative Marks | : | -2 | In all other cases. |

- *Q9. For diatomic molecules, the correct statement(s) about the molecular orbitals formed by the overlap of two $2p_z$ orbitals is(are)
- (A) σ orbital has a total of two nodal planes
- (B) σ^* orbital has one node in the xz -plane containing the molecular axis.
- (C) π orbital has one node in the plane which is perpendicular to the molecular axis and goes through the center of the molecule.
- (D) π^* orbital has one node in the xy -plane containing the molecular axis.
- Q10. The correct option(s) related to adsorption process is (are)
- (A) Chemisorption results in unimolecular layer.
- (B) The enthalpy change during physisorption is in the range of 100 to 140 kJ mol^{-1}
- (C) Chemisorption is an endothermic process
- (D) Lowering the temperature favours physisorption process
- Q11. The electrochemical extraction of aluminum from bauxite ore involves
- (A) the reaction of Al_2O_3 with coke (C) at a temperature $> 2500^\circ\text{C}$.
- (B) the neutralization of aluminate solution by passing CO_2 gas to precipitate hydrated alumina ($\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$).
- (C) the dissolution of Al_2O_3 in hot aqueous NaOH .
- (D) the electrolysis of Al_2O_3 mixed with Na_3AlF_6 to give Al and CO_2 .

- Considering the reaction sequence given below, the correct statement(s) is(are)
- $\text{H}_3\text{C}-\text{CH}_2-\text{COOH} \xrightarrow[2. \text{H}_2\text{O}]{1. \text{Br}_2, \text{red phosphorous}} \text{P} \xrightarrow[3. \text{H}_3\text{O}^+]{1. \text{Phthalimide K}^+, 2. \text{NaOH}} \text{Q} + \text{Phthalic acid}$

- Q14. Considering the following reaction sequence,


$$\text{U} = 1.\text{H}_3\text{PO}_2$$

$$2.\text{KMnO}_4.\text{KOH, heat}$$

- S =** 

- U** = 1. $\text{CH}_3\text{CH}_2\text{OH}$
2. $\text{KMnO}_4 - \text{KOH}$, heat

- T = 

SECTION 3 (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List I** and **List II**.
- **List I** has **Four** entries (I), (II), (III) and (IV) and **List II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List I** and **List II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

*Q15. Match the rate expressions in LIST-I for the decomposition of X with the corresponding profiles provided in LIST-II. X_s and k are constants having appropriate units.

LIST-I

- (I)
$$\text{rate} = \frac{k[X]}{X_s + [X]}$$

under all possible initial concentrations of X

- (II)
$$\text{rate} = \frac{k[X]}{X_s + [X]}$$

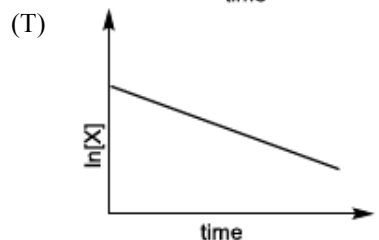
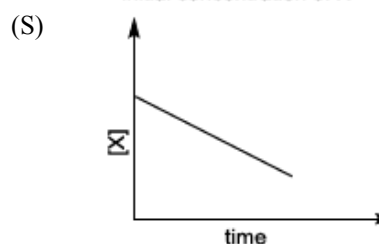
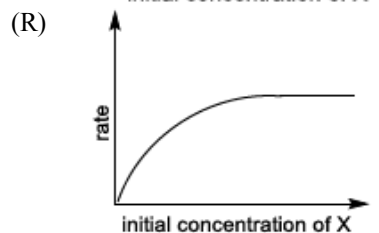
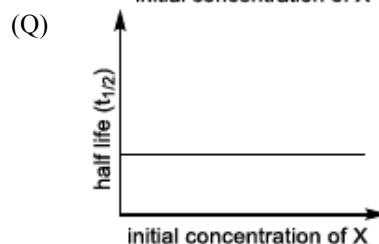
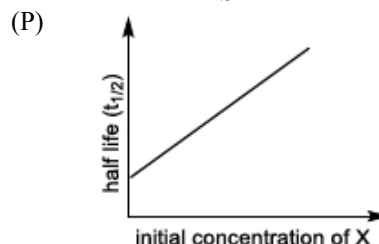
where initial concentrations of X are much less than X_s

- (III)
$$\text{rate} = \frac{k[X]}{X_s + [X]}$$

where initial concentrations of X are much higher than X_s

- (IV)
$$\text{rate} = \frac{k[X]^2}{X_s + [X]}$$

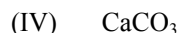
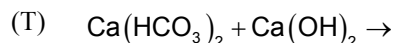
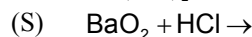
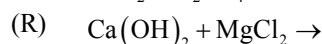
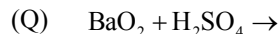
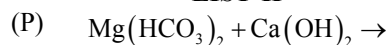
where initial concentration of X is much higher than X_s

LIST-II

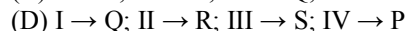
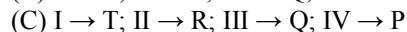
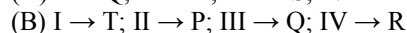
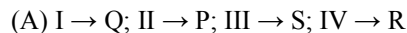
- (A) I → P; II → Q; III → S; IV → T
(C) I → P; II → Q; III → Q; IV → R

- (B) I → R; II → S; III → S; IV → T
(D) I → R; II → S; III → Q; IV → R

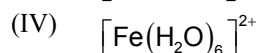
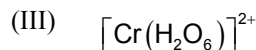
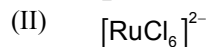
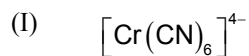
*Q16. LIST-I contains compounds and LIST-II contains reactions.

LIST-I**LIST-II**

Match each compound in LIST-I with its formation reaction(s) in LIST-II, and choose the correct option



Q17. LIST-I contains metal species and LIST-II contains their properties.

LIST-I**LIST-II**(P) t_{2g} orbitals contain 4 electrons(Q) $\mu(\text{spin-only}) = 4.9\text{BM}$

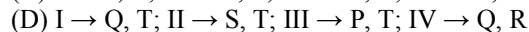
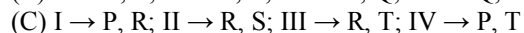
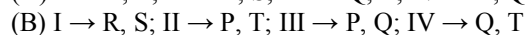
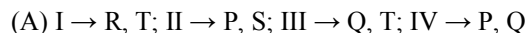
(R) Low spin complex ion

(S) Metal ion in 4+ oxidation state

(T) d^4 species

[Given: Atomic number of Cr = 24, Ru = 44, Fe = 26]

Match each metal species in LIST-I with their properties in LIST-II, and choose the correct option



Q18. Match the compounds in LIST-I with the observations in LIST-II, and choose the correct option.

LIST-I

(I) Aniline

(II) *o*-Cresol

(III) Cysteine

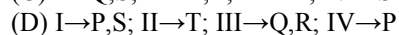
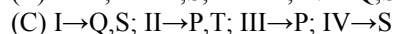
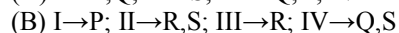
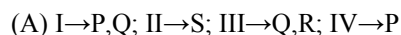
(IV) Caprolactam

LIST-II(P) Sodium fusion extract of the compound on boiling with FeSO_4 , followed by acidification with conc. H_2SO_4 , gives Prussian blue color.

(Q) Sodium fusion extract of the compound on treatment with sodium nitroprusside gives blood red color.

(R) Addition of the compound to a saturated solution of NaHCO_3 results in effervescence.

(S) The compound reacts with bromine water to give a white precipitate.

(T) Treating the compound with neutral FeCl_3 solution produces violet color.

⁶
CHEMISTRY

1.	90.39	2.	0.77	3.	10.02	4.	0.32
5.	2.38	6.	1.50	7.	1.31	8.	136.00
9.	A, D	10.	A, D	11.	B, C, D	12.	A, D
13.	B, C, D	14.	A,B,C	15.	A	16.	D
17.	A	18.	D				

CHEMISTRY

1.

$$\Delta T = 312.8 - 298 = 14.8$$

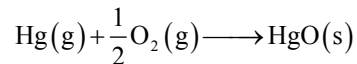
Molar heat capacity of calorimeter = 20 kJ K^{-1}

Heat released by combustion of 2 moles of Hg(g)

$$= -20 \times 14.8$$

$$= -296 \text{ kJ}$$

$$\Delta U_{\text{combustion}} = -\frac{296}{2} = -148 \text{ kJ mol}^{-1}$$



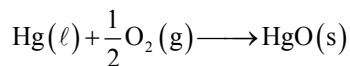
$$\Delta n_{\text{g}} = 0 - \left(1 + \frac{1}{2}\right) = -\frac{3}{2}$$

$$\Delta H_{\text{combustion}} = \Delta U_{\text{combustion}} + \Delta n_{\text{g}} RT$$

$$= -148 + \left(-\frac{3}{2}\right) \times 8.3 \times 10^{-3} \times 298$$

$$= -148 - 3.710$$

$$= -151.710 \text{ kJ mol}^{-1}$$



$$\Delta H_{\text{combustion}}^{\circ} = \Delta H_f^{\circ}(\text{HgO}) - \Delta H_{\text{Hg}(\ell) \longrightarrow \text{Hg}(\text{g})}^{\circ} + \frac{1}{2}\Delta H_f^{\circ}\text{O}_2$$

$$-151.710 = \Delta H_f^{\circ}(\text{HgO}) - 61.32 + 0$$

$$\Delta H_f^{\circ}(\text{HgO}) = -151.710 + 61.32 = 90.39$$

$$\Delta H_f^{\circ} = \mathbf{90.39}$$

2. (i) $\text{MnO}_4^{-}(\text{aq}) + 8\text{H}^{+} + 7\text{e}^{-} \longrightarrow \text{Mn}(\text{s}) + 4\text{H}_2\text{O}(\ell)$; $\Delta G_1^{\circ} = -7 \times F \times E_1^{\circ}$
 (ii) $\text{MnO}_4^{-} + 4\text{H}^{+} + 3\text{e}^{-} \longrightarrow \text{MnO}_2(\text{s}) + 2\text{H}_2\text{O}(\ell)$; $\Delta G_2^{\circ} = -3 \times F \times 1.68$
 (iii) $\text{MnO}_2(\text{s}) + 4\text{H}^{+} + 2\text{e}^{-} \longrightarrow \text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\ell)$; $\Delta G_3^{\circ} = -2 \times F \times 1.21$
 (iv) $\text{Mn}^{2+}(\text{aq}) + 2\text{e}^{-} \longrightarrow \text{Mn}(\text{s})$; $\Delta G_4^{\circ} = -2 \times F \times (-1.03)$

Now,

$$(i) = (ii) + (iii) + (iv)$$

$$\Delta G_1^{\circ} = \Delta G_2^{\circ} + \Delta G_3^{\circ} + \Delta G_4^{\circ}$$

$$-7 \times F \times E_1^{\circ} = -3 \times F \times 1.68 - 2 \times F \times 1.21 + 2 \times F \times 1.03$$

$$-7E_1^{\circ} = 3 \times 1.68 + 2 \times 1.21 - 2 \times 1.03$$

$$E_1^{\circ} = \frac{3 \times 1.68 + 2 \times 1.21 - 2 \times 1.03}{7}$$

$$E_1^{\circ} = \frac{5.04 + 2.42 - 2.06}{7}$$

$$= 0.7714$$

$$= \mathbf{0.77 \text{ V}}$$

- 3.
- | | | | | | | | |
|----------------------|-------------------------|---|---------------|-------------------|------------------|---|----------------------|
| | H_2CO_3 | + | NaOH | $\longrightarrow$ | NaHCO_3 | + | H_2O |
| Initial mol | 0.01 | | 0.01 | | 0.01 | | |
| After neutralization | 0 | | 0 | | $0.01 + 0.01$ | | |
| | | | | | $= 0.02$ | | |

Now, mixture contains 0.01 mole Na_2CO_3 and 0.02 mol of NaHCO_3 so it is a buffer.

$$\text{pH} = \text{pK}_{a_2} + \log \frac{[\text{CO}_3^{2-}]}{[\text{HCO}_3^{-}]}$$

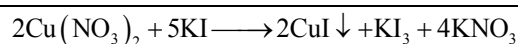
$$\text{pH} = 10.32 + \log \frac{0.01}{0.02}$$

$$= 10.32 + \log \frac{1}{2}$$

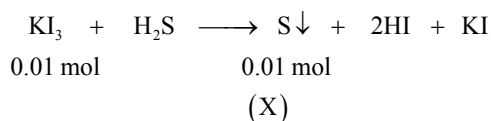
$$= 10.32 - 0.30$$

$$= 10.02$$

4. $2\text{Cu}(\text{NO}_3)_2 + 4\text{KI} \longrightarrow 2\text{CuI} \downarrow + \text{I}_2 + 4\text{KNO}_3$
 $\text{I}_2 + \text{KI} \longrightarrow \text{KI}_3$



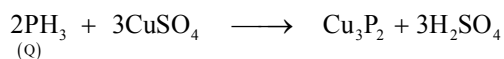
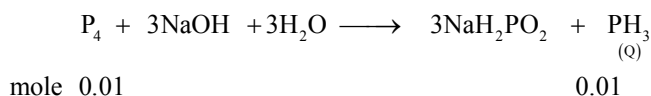
$$\text{Mole} = \frac{3.74}{187} = 0.02 \qquad \frac{0.02}{2} = 0.01$$



Mass of 'S' = $0.01 \times 32 = 0.32$

5.

Mole of $\text{P}_4 = \frac{1.24}{124} = 0.01$

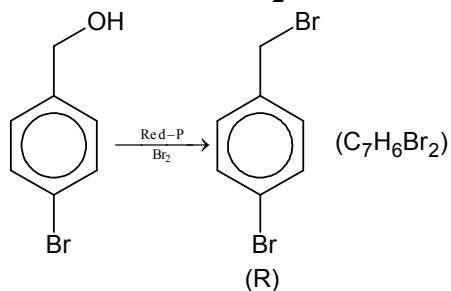


mole 0.01 $\frac{3}{2} \times 0.01$

$\therefore$ Mole of CuSO_4 required = $0.01 \times \frac{3}{2}$

Weight of $\text{CuSO}_4 = 0.01 \times \frac{3}{2} \times 159 = 2.385 \text{ g}$

6.

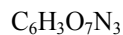
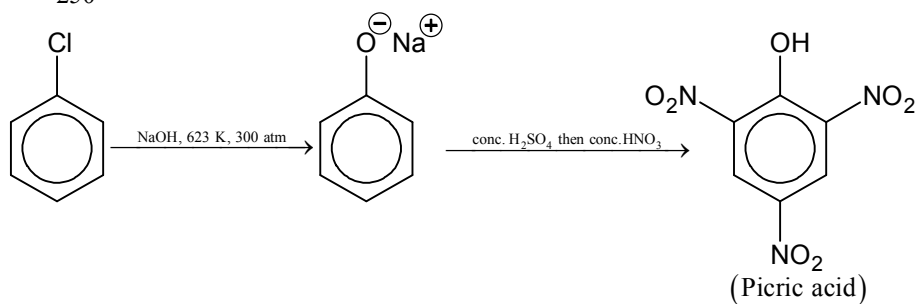


Molecular weight of (R) = $7 \times 12 + 1 \times 6 + 80 \times 2 = 250 \text{ g/mol}$

Mole of AgBr formed = $2 \times \text{mole of (R)}$

$= 2 \times \frac{1}{250} \times 188 = 8 \times 10^{-3} \times 188 = 1.504 \text{ g}$

7.

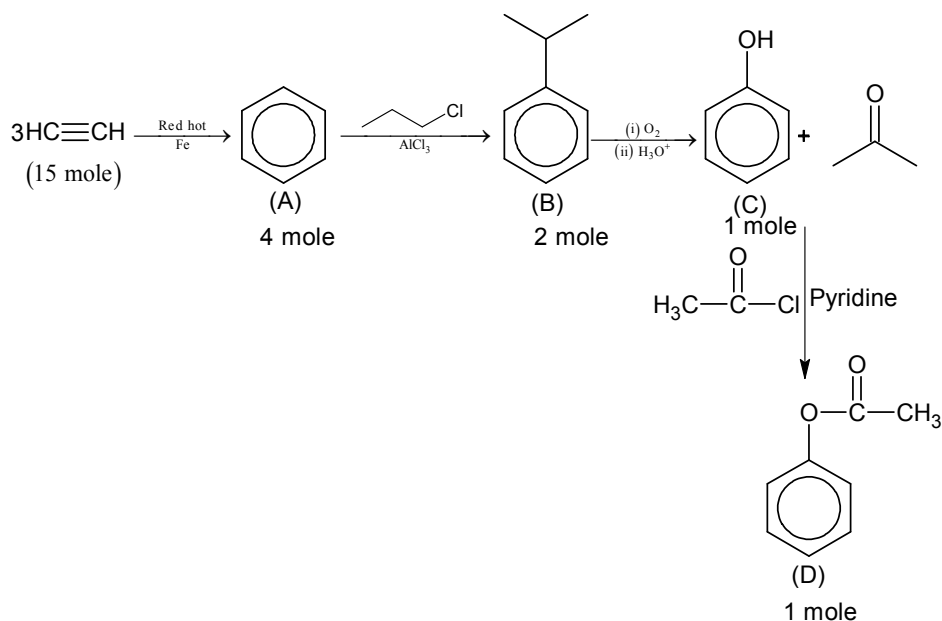


Molecular weight = $12 \times 6 + 1 \times 3 + 16 \times 7 + 14 \times 3$

$= 72 + 3 + 112 + 42 = 229 \text{ gm/mole}$

$\therefore$ Weight percentage of H = $\frac{3}{229} \times 100 = 1.3100 \%$

8.

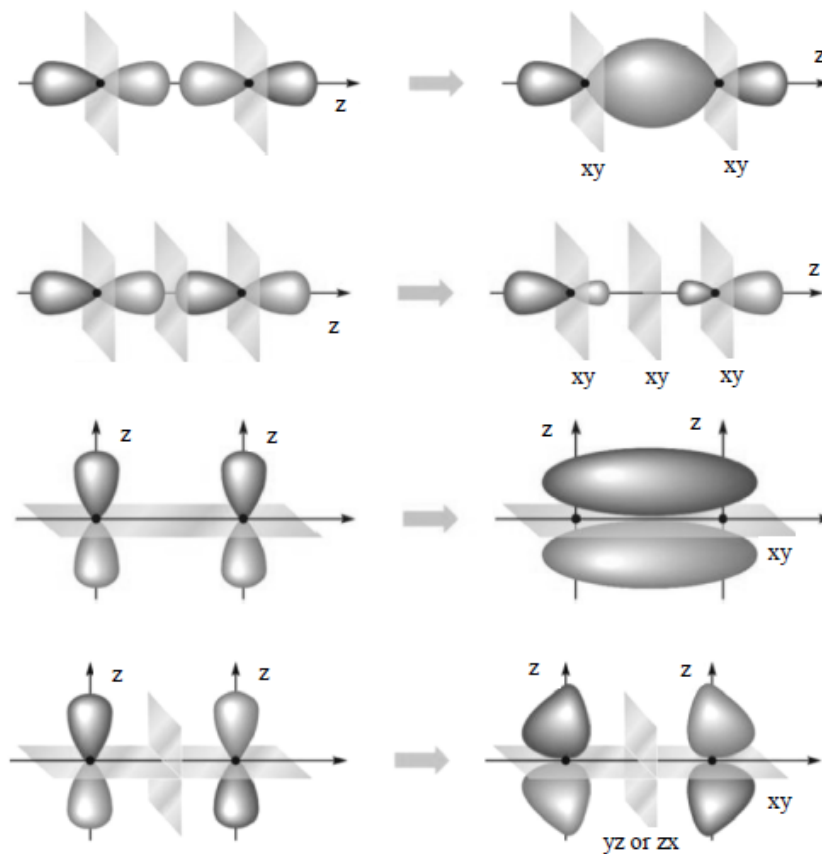
Molecular weight = $\text{C}_8\text{H}_8\text{O}_2$

$$= 12 \times 8 + 1 \times 8 + 16 \times 2 = 96 + 8 + 32 = 136$$

Moles of product (D) formed

$$= \frac{15}{3} \times 0.8 \times 0.5 \times 0.5 = \frac{3}{3} = 1 \therefore \text{Weight of product (D)} = 1 \times 136 = 136 \text{ g}$$

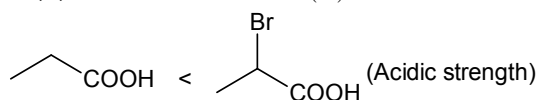
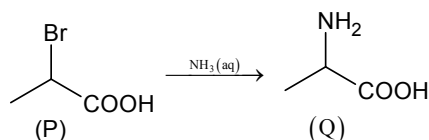
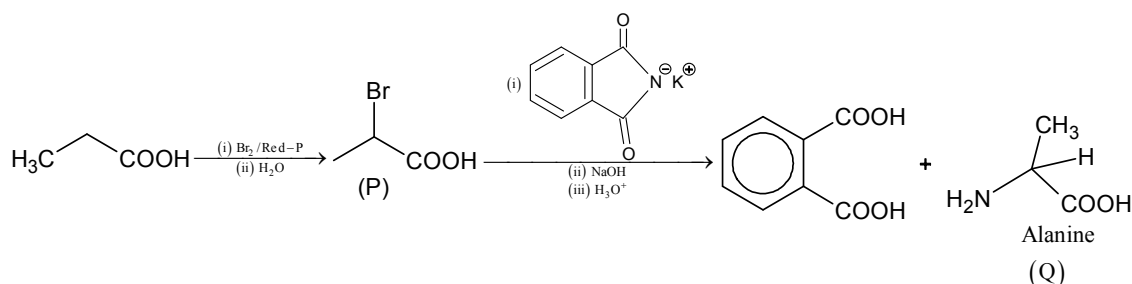
9.



10. → Chemisorption is unimolecular layer and exothermic.
 → The enthalpy change in physisorption is 20 – 40 kJ/mol.
 → Physisorption decreases with increase of temperature.
11. $2\text{NaAlO}_2 + \text{CO}_2 + 3\text{H}_2\text{O} \longrightarrow 2\text{Al}(\text{OH})_3 \downarrow + \text{Na}_2\text{CO}_3$
 $2\text{Al}(\text{OH})_3 \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 3\text{H}_2\text{O}$
 → $\text{Al}_2\text{O}_3 + 2\text{NaOH} \longrightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$
 → During electrolysis of alumina, cryolite (Na_3AlF_6) and fluorspar (CaF_2) are added to decrease the melting point of alumina.
 → Al metal is obtained at cathode while CO_2 releases at anode.

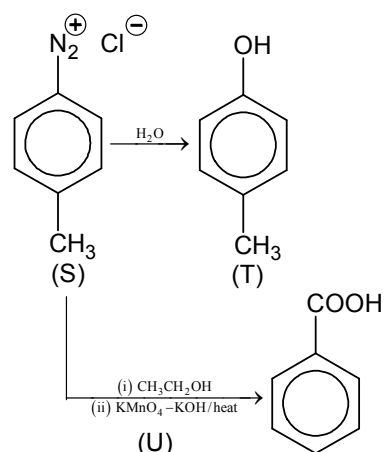
12. $3\text{PbS} + 8\text{HNO}_3 \longrightarrow 3\text{Pb}(\text{NO}_3)_2 + 3\text{S} + 2\text{NO} \uparrow + 4\text{H}_2\text{O}$
 NO → paramagnetic, neutral oxide and colourless gas

13.



14. (A)
- Reaction scheme (A):
- Starting material: 4-nitrotoluene (benzene ring with NO_2 at para and CH_3 at other para position) → (P) 4-aminotoluene (benzene ring with NH_2 at para and CH_3 at other para position) → (R) 4-diazotoluene (benzene ring with N_2^+Cl^- at para and CH_3 at other para position) → (U) Benzoic acid (benzene ring with COOH group)
- Reagents for (A): (P) to (R): H_2/Pd , ethanol; (R) to (U): (i) H_3PO_2 , (ii) KMnO_4-KOH , heat
- (B)
- Reaction scheme (B):
- Starting material: 4-nitrotoluene (benzene ring with NO_2 at para and CH_3 at other para position) → (P) 4-aminotoluene (benzene ring with NH_2 at para and CH_3 at other para position) → (R) 4-diazotoluene (benzene ring with N_2^+Cl^- at para and CH_3 at other para position)
- Reagents for (B): (P) to (R): Sn/HCl ; (R) to (S): HNO_2

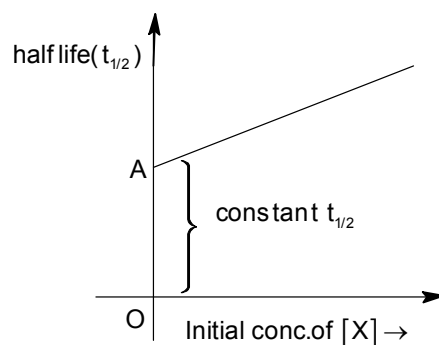
(C)



15.
$$I \rightarrow \text{Rate} = \frac{k[X]}{X_s + [X]}$$

If $[X]$ is low, then it follows first order kinetic i.e. $t_{1/2}$ is constant (OA part in graph)

If $[X]$ is high, then it follows zero order kinetics. So half life ($t_{1/2}$) varies linearly with $[X]$ as shown in graph.



I $\rightarrow$ P

II $\rightarrow \text{Rate} = \frac{k[X]}{X_s + [X]}$

If $[X]$ is less than X_s , then reaction follows Ist order kinetics as shown by the graph Q and T

III $\rightarrow \text{Rate} = \frac{k[X]}{X_s + [X]}$

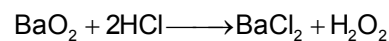
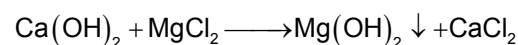
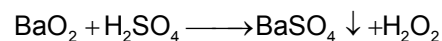
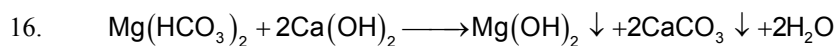
If $[X] \gg X_s$, then reaction follows zero order kinetics, as shown by the graph (S)

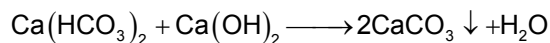
III $\rightarrow$ S

IV $\rightarrow \text{Rate} = \frac{k[X]^2}{X_s + [X]}$

If $[X] \gg [X_s]$, then reaction follows first order kinetics, as shown by the graph (Q), (T)

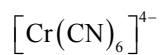
IV $\rightarrow$ Q, T



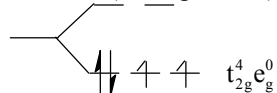
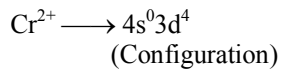


I $\rightarrow$ Q, S; II $\rightarrow$ P, R; III $\rightarrow$ S; IV $\rightarrow$ P, T

17.



I $\rightarrow$ P, T, R



Low - spin

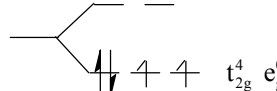
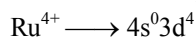
$$n = 2$$

$$\mu = \sqrt{2(2+2)}$$

$$= \sqrt{8} \text{ B.M}$$

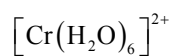


II $\rightarrow$ P, R, S,
T

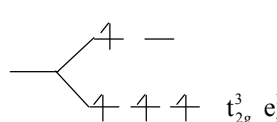
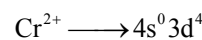


Low - spin

$$\mu = \sqrt{8} \text{ B.M}$$

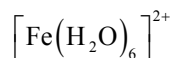


III $\rightarrow$ Q, T

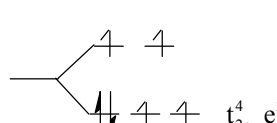
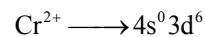


High - spin

$$\mu = \sqrt{24} \text{ B.M}$$



IV $\rightarrow$ P, T

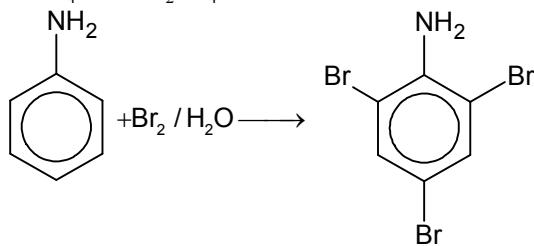


High - spin

$$\mu = \sqrt{4(4+2)} = \sqrt{24} \text{ B.M}$$

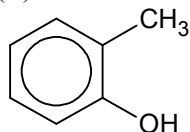
18.

(I) since aniline contains C and N so its sodium extract will give Prussian Blue colour with $\text{FeSO}_4/\text{Conc. H}_2\text{SO}_4$



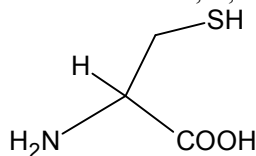
white ppt

(II) Produces violet colour with neutral FeCl_3



(III) It contains $-\text{COOH}$, so it give effervescence with NaHCO_3

It also contains C, N, S, so its Na-extract will give blood red colour with $\text{FeCl}_3/\text{H}_2\text{SO}_4$.



(IV) Caprolectum, it contains C and N, so its sodium extract will give Prussian blue colour

